Vivek kaklotar

Module 2 – Introduction to Programming

Q-1 Overview of C Programming

THEORY EXERCISE: Write an essay covering the history and evolution of C programming. Explain its importance and why it is still used today.

C programming language was developed in the 1970s by Dennis Ritchie at Bell Labs. It was created as an improved version of the B language and mainly used to develop the UNIX operating system. Later, standard versions like ANSI C (1989), C99, and C11 were introduced.

Importance of C

C is a powerful and basic programming language. It is fast, efficient, and works close to hardware. Many modern languages like C++ and Java are influenced by C, so it is called the "mother language."

Why C is Still Used Today

C is still used because it is fast, portable, and efficient. It is widely used in system programming, embedded systems, and applications where performance is very important.

LAB EXERCISE: Research and provide three real-world applications where C programming is extensively used, such as in embedded systems, operating systems, or game development.

1. Embedded Systems
   - C is used in devices like washing machines, microwaves, and IoT systems because it can directly control hardware and uses less memory.
2. Operating Systems
   - Operating systems like Linux and UNIX are mostly written in C because it provides fast execution and better memory management.
3. Game Development
   - C is used in game development for high performance and speed, especially in game engines and graphics processing.

Q-2 **Setting Up Environment**

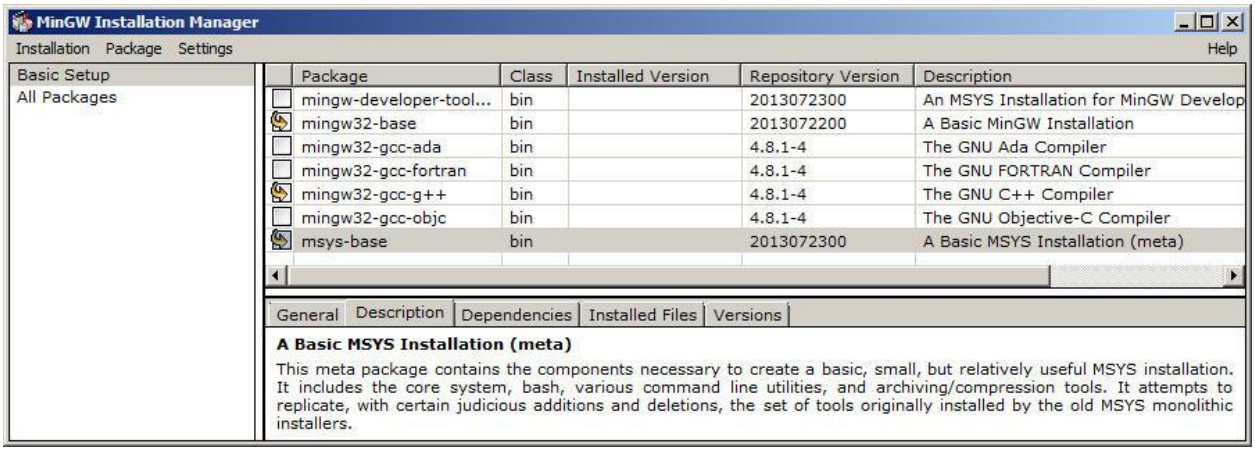
THEORY EXERCISE: Describe the steps to install a C compiler (e.g., GCC) and set up an Integrated Development Environment (IDE) like DevC++, VS Code, or CodeBlocks.

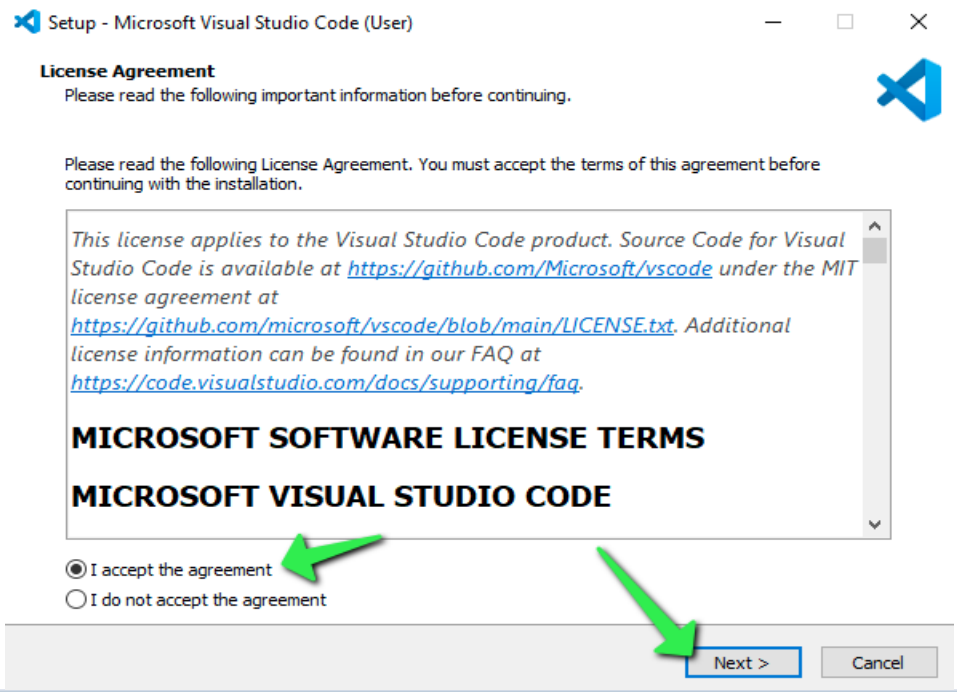
Steps to Install C Compiler and IDE

1. Download Compiler (GCC)
   Install GCC compiler using MinGW (Windows) or directly on Linux/Mac.
2. Set Environment Path
   Add the compiler path (bin folder) to system environment variables.
3. Install IDE
   Download and install any IDE like Dev-C++, CodeBlocks, or VS Code.
4. Configure IDE
   Link the IDE with GCC compiler (select compiler path in settings).
5. Test Installation
   Write a simple program and compile it to check if everything works properly.

LAB EXERCISE: Install a C compiler on your system and configure the IDE. Write your first program to print "Hello, World!" and run it.

1. Install Compiler and IDE




Setup - Microsoft Visual Studio Code (User)
License Agreement
Please read the following important information before continuing.
Please read the following License Agreement. You must accept the terms of this agreement before continuing with the installation.
This license applies to the Visual Studio Code product. Source Code for Visual Studio Code is available at https://github.com/Microsoft/vscode under the MIT license agreement at https://github.com/microsoft/vscode/blob/main/LICENSE.txt. Additional license information can be found in our FAQ at https://code.visualstudio.com/docs/supporting/faq.
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I accept the agreement
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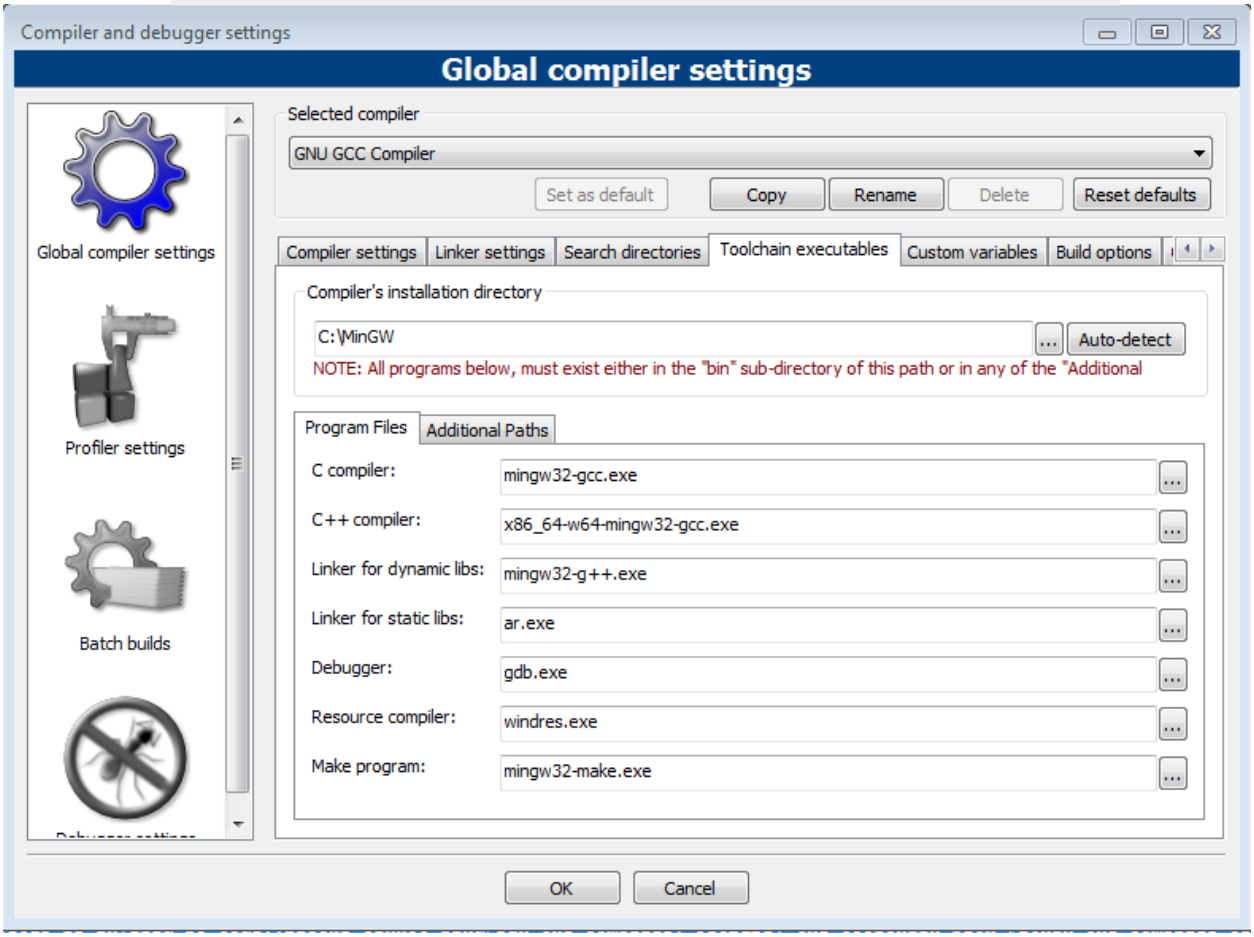
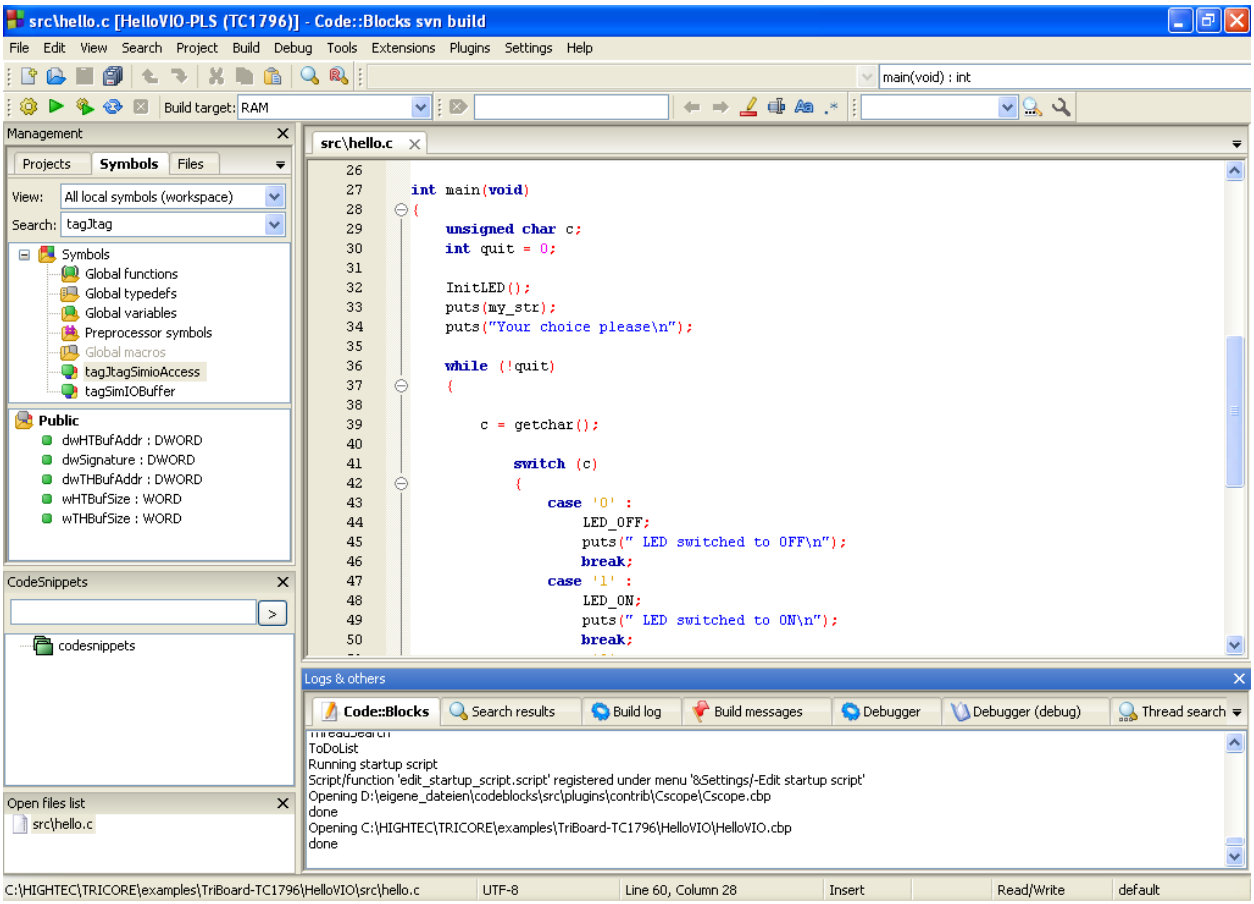

Compiler and debugger settings
Global compiler settings
Global compiler settings
Profiler settings
Batch builds
Selected compiler
GNU GCC Compiler
Set as default
Copy
Rename
Delete
Reset defaults
Compiler settings
Linker settings
Search directories
Toolchain executables
Custom variables
Build options
Compiler's installation directory
C:\MinGW
Auto-detect
NOTE: All programs below, must exist either in the "bin" sub-directory of this path or in any of the "Additional
Program Files
Additional Paths
C compiler:
mingw32-gcc.exe
C++ compiler:
x86_64-w64-mingw32-gcc.exe
Linker for dynamic libs:
mingw32-g++.exe
Linker for static libs:
ar.exe
Debugger:
gdb.exe
Resource compiler:
windres.exe
Make program:
mingw32-make.exe
OK
Cancel



Install MinGW (GCC compiler)
Install VS Code or CodeBlocks
Configure compiler in IDE

2. Write First Program

```
#include <stdio.h>
int main() {
    printf("Hello, World!");
    return 0;
}
```

3. Compile and Run
   - Save file as `hello.c`
   - Compile using `gcc hello.c`
   - Run using `./a.out` or hello.exe

Q-3 Basic Structure of a C Program

THEORY EXERCISE: Explain the basic structure of a C program, including headers, main function, comments, data types, and variables. Provide examples.

C program has the following main parts:

1. **Header Files**
   Used to include libraries.
   Example: `#include <stdio.h>`
2. **Main Function**
   Execution of program starts from `main()` function.
   Example:
   ```
   int main() {
       // code
   }
   ```
3. **Comments**
   Used to explain code.
   - Single line: `// comment`
   - Multi-line: `/* comment */`
4. **Data Types**
   Define type of data.
   Examples:
   - `int` → integer
   - `float` → decimal number
   - `char` → character
5. **Variables**
   Used to store values.
   Example:
   ```
   int a = 10;
   float b = 5.5;
   char c = 'A';
   ```

LAB EXERCISE: Write a C program that includes variables, constants, and comments. Declare and use different data types (int, char, float) and display their values.

```
#include <stdio.h>

// This program shows variables, constants, and data types

int main()
{
    // Variable declaration
    int age = 20;
    float height = 5.8;
    char grade = 'A';

    // Constant declaration
    const int year = 2026;

    // Display values
    printf("Age: %d\n", age);
    printf("Height: %.1f\n", height);
    printf("Grade: %c\n", grade);
    printf("Year: %d\n", year);

    return 0;
}
```

Q-4 Operators in C

THEORY EXERCISE: Write notes explaining each type of operator in C: arithmetic, relational, logical, assignment, increment/decrement, bitwise, and conditional operators..

1. **Arithmetic Operators**
   Used for mathematical calculations.
   Examples: `+ , - , * , / , %`
2. **Relational Operators**
   Used to compare values.
   Examples: `== , != , > , < , >= , <=`
   Result is true (1) or false (0)
3. **Logical Operators**
   Used to combine conditions.
   Examples: `&& (AND), || (OR), ! (NOT)`
4. **Assignment Operators**
   Used to assign values to variables.
   Examples: `= , += , -= , *= , /=`
5. **Increment/Decrement Operators**
   Used to increase or decrease value by 1.
   Examples: `++ , --`
6. **Bitwise Operators**
   Work on binary values (bits).
   Examples: `& , | , ^ , ~ , << , >>`
7. **Conditional Operator (Ternary)**
   Short form of if-else.
   Example:
   ```
   (condition) ? true_value : false_value;
   ```

LAB EXERCISE: Write a C program that accepts two integers from the user and performs arithmetic, relational, and logical operations on them. Display the results.

```
#include <stdio.h>

int main() {
    int a, b;

    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);

    // Arithmetic operations
    printf("Addition: %d\n", a + b);
    printf("Subtraction: %d\n", a - b);
    printf("Multiplication: %d\n", a * b);
    printf("Division: %d\n", a / b);

    // Relational operations
    printf("a > b: %d\n", a > b);
    printf("a == b: %d\n", a == b);
    printf("a != b: %d\n", a != b);

    // Logical operations
    printf("(a > 0 && b > 0): %d\n", (a > 0 && b > 0));
    printf("(a > 0 || b > 0): %d\n", (a > 0 || b > 0));
    printf("!(a > b): %d\n", !(a > b));

    return 0;
}
```

Q-5 Control Flow Statements in C

THEORY EXERCISE: Explain decision-making statements in C (if, else, nested if-else, switch). Provide examples of each.

Decision-Making Statements in C

1. if Statement
   Used to execute code when a condition is true.

```
if (a > b) {
    printf("A is greater");
}
```

2. if-else Statement
   Used when there are two choices.

```
if (a > b) {
    printf("A is greater");
} else {
    printf("B is greater");
}
```

3. Nested if-else
   if-else inside another if-else for multiple conditions.

```
if (a > b) {
    if (a > c) {
        printf("A is largest");
    }
}
```

4. switch Statement
   Used to select one option from many.

```
switch (num) {
    case 1: printf("One"); break;
    case 2: printf("Two"); break;
    default: printf("Invalid");
}
```

LAB EXERCISE: Write a C program to check if a number is even or odd using an if-else statement. Extend the program using a switch statement to display the month name based on the user's input (1 for January, 2 for February, etc.)

```
#include <stdio.h>

int main() {
    int num, month;

    // Even or Odd
    printf("Enter a number: ");
    scanf("%d", &num);

    if (num % 2 == 0) {
        printf("Number is Even\n");
    } else {
        printf("Number is Odd\n");
    }

    // Month using switch
    printf("Enter month number (1-12): ");
    scanf("%d", &month);

    switch (month) {
        case 1: printf("January"); break;
        case 2: printf("February"); break;
        case 3: printf("March"); break;
        case 4: printf("April"); break;
        case 5: printf("May"); break;
        case 6: printf("June"); break;
        case 7: printf("July"); break;
        case 8: printf("August"); break;
        case 9: printf("September"); break;
        case 10: printf("October"); break;
        case 11: printf("November"); break;
        case 12: printf("December"); break;
        default: printf("Invalid month");
    }

    return 0;
}
```

Q-6 Looping in C

THEORY EXERCISE: Compare and contrast while loops, for loops, and do-while loops. Explain the scenarios in which each loop is most appropriate.

Comparison of Loops

1. **while Loop**
   - Checks condition before execution
   - Used when number of iterations is not known

```
while (condition) {
    // code
}
```

2. **for Loop**
   - Used when number of iterations is known
   - Initialization, condition, update in one line

```
for (i = 1; i <= 10; i++) {
    // code
}
```

3. **do-while Loop**
   - Executes at least once (condition checked after)
   - Useful when loop must run at least one time

```
do {
    // code
} while (condition);
```

LAB EXERCISE: Write a C program to print numbers from 1 to 10 using all three types of loops (while, for, do-while).

```
#include <stdio.h>

int main() {
    int i;

    // while loop
    i = 1;
    while (i <= 10) {
        printf("%d ", i);
        i++;
    }

    printf("\n");

    // for loop
    for (i = 1; i <= 10; i++) {
        printf("%d ", i);
    }

    printf("\n");

    // do-while loop
    i = 1;
    do {
        printf("%d ", i);
        i++;
    } while (i <= 10);

    return 0;
}
```

Q-7 Loop Control Statements

THEORY EXERCISE: Explain the use of break, continue, and goto statements in C. Provide examples of each.

1. **break Statement**
   - Used to exit the loop immediately

```
if (i == 5) {
    break;
}
```

2. **continue Statement**
   - Skips current iteration and moves to next

```
if (i == 3) {
    continue;
}
```

3. **goto Statement**
   - Transfers control to a labeled statement

```
goto label;
label:
printf("Hello");
```

LAB EXERCISE: Write a C program that uses the break statement to stop printing numbers when it reaches 5. Modify the program to skip printing the number 3 using the continue statement.

```
#include <stdio.h>
int main() {
    int i;

    // Using break
    for (i = 1; i <= 10; i++) {
        if (i == 5) {
            break;
        }
        printf("%d ", i);
    }

    printf("\n");

    // Using continue
    for (i = 1; i <= 10; i++) {
        if (i == 3) {
            continue;
        }
        printf("%d ", i);
    }

    return 0;
}
```

Q-8. Functions in C

THEORY EXERCISE: What are functions in C? Explain function declaration, definition, and how to call a function. Provide examples.

Functions are blocks of code used to perform a specific task and improve reusability.

- Declaration → tells function name and parameters
- Definition → actual code of function
- Call → executes the function

```
int add(int a, int b); // declaration

int add(int a, int b) { // definition
    return a + b;
}

int main() {
    int sum = add(2, 3); // call
}
```

LAB EXERCISE: Write a C program that calculates the factorial of a number using a function. Include function declaration, definition, and call.

```
#include <stdio.h>

int fact(int n); // declaration

int fact(int n) { // definition
    int i, f = 1;
    for(i = 1; i <= n; i++)
        f *= i;
    return f;
}

int main() {
    int num;
    printf("Enter number: ");
    scanf("%d", &num);

    printf("Factorial = %d", fact(num)); // call
    return 0;
}
```

Q-9. Arrays in C

THEORY EXERCISE: Explain the concept of arrays in C. Differentiate between one-dimensional and multi-dimensional arrays with examples.

Array = collection of same type elements.

1D Array

```
int a[5] = {1,2,3,4,5};
```

2D Array

```
int a[3][3];
```

Difference:

- 1D → single row
- 2D → rows + columns (matrix)

LAB EXERCISE: Write a C program that stores 5 integers in a one-dimensional array and prints them. Extend this to handle a two-dimensional array (3x3 matrix) and calculate the sum of all elements.

```
#include <stdio.h>

int main() {
    int i, j, sum = 0;
    int a[5] = {1,2,3,4,5};

    // 1D array print
    for(i = 0; i < 5; i++)
        printf("%d ", a[i]);

    printf("\n");

    // 2D array
    int b[3][3] = {
        {1,2,3},
        {4,5,6},
        {7,8,9}
    };

    for(i = 0; i < 3; i++) {
        for(j = 0; j < 3; j++) {
            sum += b[i][j];
        }
    }

    printf("Sum = %d", sum);
    return 0;
}
```

Q-10. Pointers in C

THEORY EXERCISE: Explain what pointers are in C and how they are declared and initialized. Why are pointers important in C?

Pointers store the address of a variable.

```
int a = 10;
int *p = &a;
```

Importance:
Direct memory access
Used in arrays, functions, dynamic memory

LAB EXERCISE: Write a C program to demonstrate pointer usage. Use a pointer to modify the value of a variable and print the result.

```
#include <stdio.h>

int main() {
    int a = 10;
    int *p = &a;

    *p = 20; // modify using pointer

    printf("Value of a = %d", a);
    return 0;
}
```

Q-11. Strings in C

THEORY EXERCISE: Explain string handling functions like strlen(), strcpy(), strcat(), strcmp(), and strchr(). Provide examples of when these functions are useful.

`strlen()` → length
`strcpy()` → copy
`strcat()` → join strings
`strcmp()` → compare
`strchr()` → find character.

LAB EXERCISE: Write a C program that takes two strings from the user and concatenates them using strcat(). Display the concatenated string and its length using strlen().

```
#include <stdio.h>
#include <string.h>

int main() {
    char s1[50], s2[50];

    printf("Enter first string: ");
    scanf("%s", s1);

    printf("Enter second string: ");
    scanf("%s", s2);

    strcat(s1, s2);

    printf("Concatenated string: %s\n", s1);
    printf("Length: %lu", strlen(s1));

    return 0;
}
```

Q-12. Structures in C

THEORY EXERCISE: Explain the concept of structures in C. Describe how to declare, initialize, and access structure members.

Structure = collection of different data types.

```
struct Student {
    char name[20];
    int roll;
    float marks;
};
```

LAB EXERCISE: Write a C program that defines a structure to store a student's details (name, roll number, and marks). Use an array of structures to store details of 3 students and print them.

```
#include <stdio.h>

struct Student {
    char name[20];
    int roll;
    float marks;
};

int main() {
    struct Student s[3];

    for(int i=0;i<3;i++) {
        printf("Enter name, roll, marks: ");
        scanf("%s %d %f", s[i].name, &s[i].roll, &s[i].marks);
    }

    for(int i=0;i<3;i++) {
        printf("%s %d %.2f\n", s[i].name, s[i].roll, s[i].marks);
    }

    return 0;
}
```

Q-13. File Handling in C

THEORY EXERCISE: Explain the importance of file handling in C. Discuss how to perform file operations like opening, closing, reading, and writing files. .

`fopen()` → open file
`fclose()` → close file
`fprintf()` → write
`fscanf()` → read

LAB EXERCISE: Write a C program to create a file, write a string into it, close the file, then open the file again to read and display its contents.

```
#include <stdio.h>

int main() {
    FILE *fp;

    // write
    fp = fopen("test.txt", "w");
    fprintf(fp, "Hello World");
    fclose(fp);

    // read
    char str[50];
    fp = fopen("test.txt", "r");
    fscanf(fp, "%s", str);
    printf("Data: %s", str);
    fclose(fp);

    return 0;
}
```